Then test Mininet, as follows:

```
Sudo mn  -test pingall
```

The Mininet command cheat sheet

1. The following command runs the default minimal topology which includes the OpenFlow kernel switch connected to two hosts with the reference controller:

```
Mininet> sudo mn
```

2. The following command is used to get the help of any command in Mininet:

```
Mininet>help
```

3. The following command displays the nodes of the network:

```
Mininet>nodes
```

4. The following displays the links of a network:

```
 Mininet> net
```

5. To dump information about all nodes, use the command below:

```
Mininet> dump
```

6. To test connectivity between hosts, you can ping from Host 0 to Host 1, as follows:

```
Mininet> h1 ping -c1 h2
```

7. The following command will start the two separate terminals for node h1 and h2:

```
Mininet> xterm h1 h2
```

8. The command below is used to open the Wireshark network packet analyser tool to monitor and analyse the behaviour of incoming and outgoing packets:

```
Mininet> wireshark &
```

9. A few performance modeling commands in Mininet:

The following command establishes the transmission control link with CPU limited hosts:

```
net = Mininet(link=TCLink, host=CPULimitedHost)
# Limit link bandwidth and add delay
```

The following command is used to set the link's bandwidth and delay timing:

```
net.addLink(h2, s1, bw=10, delay='50ms')
```

10. To exit from the command line interface (CLI), use the following command:

```
Mininet> exit
```

11. To clean up the Mininet network configuration, issue the following command:

```
Mininet> sudo mn -c
```

Creating a network

Mininet supports parametrised topologies. You can create a flexible custom topology (in Python, with just a few lines of code) which can be configured based on the parameters you pass onto it, and this can be reused for many experiments. You can also set the network's performance and 'quality of service' parameters.

You can create a network topology as shown in Figure 1 using Mininet Python code.

```
Mininet code
net = Mininet()            # net is a Mininet() object
h1 = net.addHost( 'h1' )      # h1 is a Host() object
h2 = net.addHost( 'h2' )      # h2 is a Host()
s1 = net.addSwitch( 's1' )    # s1 is a Switch() object
c0 = net.addController( 'c0' )    # c0 is a Controller()
net.addLink( h1, s1 )     # creates a Link() object
net.addLink( h2, s1 )
net.start()
CLI( net )
net.stop()
```

The above code creates a topology with two hosts, one switch and remote controller. In the example shown above, h1 and h2 are the two hosts connected to controller c0 through the virtual switch s1. I will give more details on this in the section on simulation.

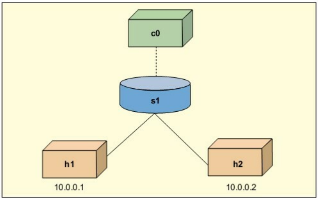

Figure 1: Network topology